Fiscal Year:	FY 2024	**Task Last Updated:**	FY 02/26/2025
PI Name:	Davila, Alfonso Ph.D.		
Project Title:	Center for Life Detection Research and Service (CLD-RS) -- Computational Systems Biology at the Ecosystem Scale		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	NASA Ames Research Center		
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City:	Moffett Field	**State:**	CA
Zip Code:	94035	**Congressional District:**	16
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	10/01/2022	**End Date:**	09/30/2024
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	NOTE: Per NASA Ames Research Center (ARC), the original Principal Investigator (PI) for this project was Andrzej Pohorille, Ph.D. However, Dr. Pohorille has passed away. Alfonso Davila, Ph.D. is the new PI for this investigation and is providing the final reporting/closeout paperwork (Ed., 2/26/25).		
COI Name (Institution):	Broddrick, Jared T. Ph.D. (NASA Ames Research Center) Everroad, R. Craig Ph.D. (NASA Ames Research Center)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The purpose of CLD/RS is to support the planning and implementation of missions that will seek evidence of life beyond Earth. We propose to do so by: conducting research on “detectability” to help define target/sample selection and measurement strategies/requirements (Task 1); developing tools and engagement activities that enable members of the broader astrobiology community to formulate their knowledge, research, and expertise in a way that facilitates use in mission planning (Task 2); and supporting the broader technology community in maturing promising measurement approaches toward potential use in flight (Task 3). The work is directly responsive to recommendations in the National Academies of Sciences, Engineering, and Medicine (NASEM) consensus study report on astrobiology strategy that NASA should: “support expanding biosignature research to address gaps in understanding biosignature preservation and the breadth of possible false positives and false negative signatures” (Task 1); “aid the community in developing a comprehensive framework…to guide testing and evaluation of in situ and remote biosignatures” (Task 2); and “accelerate the development and validation, in relevant environments, of mission-ready, life detection technologies” (Task 3).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The work is directly responsive to recommendations in the National Academies of Sciences, Engineering, and Medicine (NASEM) consensus study report on astrobiology strategy that NASA should: “support expanding biosignature research to address gaps in understanding biosignature preservation and the breadth of possible false positives and false negative signatures” (Task 1); “aid the community in developing a comprehensive framework…to guide testing and evaluation of in situ and remote biosignatures” (Task 2); and “accelerate the development and validation, in relevant environments, of mission-ready, life detection technologies” (Task 3).
Task Progress:	The objective of this project was to develop a framework that constrains biosignature detectability as a function of spacecraft and/or telescope observables, with applications to Mars. To achieve our objective, we characterized the influence of environmental factors on biological potential at a microbial physiological level, through a combination of experimental, field, and theoretical approaches designed to develop and validate a predictive modeling capability. Overall progress/ accomplishments/ problems. 1. Climate Data Analysis: We successfully compiled and analyzed climate data to define a humidity gradient across the Atacama Desert. Environmental sensors (HOBO Onset RH/T data loggers) previously deployed in the region recorded temperature and relative humidity over time. These data provided a detailed understanding of local climatic conditions and served as a foundation for interpreting microbial colonization patterns. 2. Soil Water Availability: To estimate water availability, we measured soil evaporation rates across different substrates in the Atacama Desert. This was achieved using an electrical scale capable of data logging. The results offered insights into how substrate type influences evaporation rates, which in turn affects microbial habitability. 3. Development of Methods for GEM Parameterization: We established cultures and developed methods for parameterizing the Genome-Scale Metabolic Model (GEM). This involved collecting data necessary for simulating primary production in abiotic substrates. The oxygen evolution rate, a critical parameter for GEM simulations, was shared with the PI team working on the Space Algae II project under NASA’s BioProtection System (BPS). 4. Space Algae Collaboration: In collaboration with the BPS team, we measured oxygen production rates for a spirulina strain that will soon be cultivated on the International Space Station (ISS). These measurements are pivotal for understanding how this cyanobacterium adapts to space environments, potentially supporting life-support systems in extraterrestrial habitats. 5. Fieldwork in the Atacama Desert: During fieldwork conducted in April and May 2024, we visited multiple sites exhibiting distinct microbial colonization modes, including soil and salt substrates. Halite nodules were collected for laboratory analysis to explore microbial communities within these substrates. Additionally, environmental data downloaded from sensors allowed us to examine the impact of a significant rainfall event on soil microbial communities. 6. Acquisition of Dynamic Environmental Chamber: We procured a dynamic environmental chamber designed to simulate desert conditions in a controlled laboratory setting. Although delays in installation due to administrative hurdles and facility breakdowns postponed its use, this chamber will be integral to future experiments in FY25 and beyond. 7. Student Mentorship: Two students were mentored as part of the project: Ellen Cook conducted experiments to measure acclimated growth rates in Synechococcus elongatus PCC7942 across a light gradient from inhibiting to limiting levels. She also collected chlorophyll a (Chl a) and oxygen evolution data, which contributed to model parameterization. Cesar Meza explored the concept of a minimum viable population (MVP) in bacteria. Using Synechococcus elongatus PCC7942, he designed experiments with nutrient-replete and nutrient-poor media, exposing cultures to UV light as a stressor. Overall, this project made substantial progress in advancing our understanding of microbial ecology and environmental interactions in the Atacama Desert. From analyzing climate data to conducting field and laboratory experiments, the findings offer valuable insights into life’s adaptability under extreme conditions. Despite some logistical setbacks, the groundwork laid by this research will support ongoing and future studies, including those involving space-based applications.
Bibliography Type:	Description: (Last Updated: 07/08/2026)
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